# Nathaniel Bottman

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Machine learning researcher working on reinforcement-learning post-training and evaluation of large language models (LLMs): when training teaches a model to report something other than what it has computed, and label-free ways to detect when its reasoning fails. PI of a DARPA-funded research program ("An Operad-Based Framework for Deep Learning," $836K). Introduced *operadic consistency*, a label-free signal that checks a model's answer against its own decomposition of the problem. Previously a research group leader at the Max Planck Institute for Mathematics.

## Experience

**Incubilate** – Founder & Principal Scientist Seattle, WA, July 2025–present

- PI of "An Operad-Based Framework for Deep Learning" ($836K, DARPA AIQ program): operads as the mathematical structure of question decomposition, and label-free grading of multi-step reasoning.
- Showed when verifiable-reward RL teaches a model to bluff (preprint, Nov 2026, with A. Oblomkov). Derived a first-order condition under which a GRPO step makes the model more honest, and tested it with GRPO runs on propositional Datalog (3B), two code models, and abstention on grade-school math.
  - No run above the predicted threshold ends collapsed. Near it, honesty becomes a race that a checker can win and a lottery with the same starting payoff cannot; a model of the race, fit only on runs outside the race band, predicted held-out outcomes before those runs were done.
  - Just above the threshold, runs first dip toward bluffing (on code nearly to zero) and then recover; far above it, the model starts giving the honest verdict where it is wrong (both code models call about half of correct programs buggy at $pr/b = 3$).
  - A generous-then-cut reward schedule keeps a depth-3 Datalog model at least 67% honest in three seeds, at a ratio where the same model trained from the start falls to 4–15%. When honesty collapses, a linear probe fit before RL still reads the true verdict.
- Developed *operadic consistency* (OC): tracks accuracy at $r = 0.86$–$0.94$ across twelve LLMs on four multi-hop QA benchmarks, predicts a held-out model's accuracy to 3.0 pp with no labels, and improves selective prediction over a tuned self-consistency baseline at equal cost.
- Extended OC to frontier reasoning models (DeepSeek-R1, Kimi-K2.6, GLM-5.1) by extracting the decomposition from the model's own chain of thought and re-deriving it; positive lift on all 16 selective-prediction cells tested.
- Integrated OC as an accuracy predictor in a DARPA evaluation harness, incl. a trace-extracted OC node for reasoning models; stack spans open-weights (4B–671B) and API models.

**Max Planck Institute for Mathematics** – Research Group Leader (W2) Bonn, Jan 2021–Jun 2025

- Led a research group in symplectic geometry; supervised six postdoctoral researchers and multiple student theses; served on the institute's Scientific Committee (hiring).

**Postdoc positions:** U. Southern California (2019–20), Inst. for Advanced Study (2016–19), Northeastern (2015–16).

## Selected Publications

1. **Honesty is a race: predicting when verifiable-reward RL teaches a model to bluff.** N. Bottman, A. Oblomkov. Preprint, 2026.
   Blog post: `natebottman.com/blog/honesty-is-a-race`.
2. **Operadic consistency: a label-free signal for compositional reasoning failures in LLMs.** N. Bottman, Y. Liu, K. Richardson. arXiv:2606.13649. NeurIPS 2026 (poster).
   Code: `github.com/natebottman/operadic-consistency-paper`.
3. **Operads for compositional reasoning in LLMs.** N. Bottman, K. Richardson. arXiv:2606.13634. Presented at the Compositional Learning and the Combining Theory and Benchmarks workshops, ICML 2026.
4. **How regularization affects the geometry of loss functions.** N. Bottman, Y. Cooper, A. Lerario. arXiv:2307.15744.

Nineteen further publications in symplectic geometry, combinatorics, and applied mathematics, in venues including the *Bulletin of the AMS*, *Selecta Mathematica*, *Algebraic & Geometric Topology*, and *J. Symplectic Geometry*. These include a polytopal realization of higher-categorical associahedra found with OpenAI's Astra model and formalized in Lean 4 (arXiv:2609.25316).

## Grants & Fellowships

- DARPA AIQ program, "An Operad-Based Framework for Deep Learning," $836K, 2025–present (PI).
- NSF Standard Grant DMS-1906220, $144K, 2019–2022 (PI).
- NSF Mathematical Sciences Postdoctoral Research Fellowship, 2016–2019.
- NSF Graduate Research Fellowship, 2010–2015.

## Education

**Massachusetts Institute of Technology**, Ph.D., Mathematics, 2015. Advisor: Katrin Wehrheim.
**University of Washington**, B.S. Mathematics and B.A. Slavic Languages & Literatures, 2010. Dean's Medal (top graduating senior in the natural sciences).

## Service & Community

Co-organizer, Combining Theory and Benchmarks Workshop, ICML 2026 (Seoul); co-organizer of research seminars and workshops since 2012, incl. the Western Hemisphere Virtual Symplectic Seminar (2020–23). 45+ invited talks.

## Technical

Python (GRPO post-training and RL experiments; LLM evaluation pipelines; linear probes; open-weights and API inference), PyTorch, LaTeX.